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UPI Transaction Analytics Dashboard | Microsoft Excel

Interactive UPI Transaction Analysis Using Microsoft Excel

1. Introduction

Unified Payments Interface (UPI) has become an important digital payment mechanism, enabling users to transfer money and make payments through mobile applications. With a large number of transactions being generated across different customers, locations, banks, payment applications and transaction types, analyzing transaction data can help identify important patterns and trends.

This project focuses on analyzing a sample dataset containing **200,000 UPI transactions from May 2026** using Microsoft Excel. The objective is to transform transaction-level data into meaningful business insights through data preparation, calculations, PivotTables, slicers and interactive dashboards.

Two interactive dashboard pages were developed to analyze transaction amount, customer demographics, geographical distribution, banking channels, transaction types, UPI applications, transaction status and time-based patterns.

Figure 1: Sample of the UPI Transactions Dataset

Transaction ID	Transaction Ref	Day	Transaction Time	Hour	UPI App	Customer	Age Group	Gender	State	City	Merchant Name	Merchant Category	Transaction Type	Payment Mode	Bank Name	Amount (₹)	Cashback (₹)	Transaction Fee (₹)	Status	Failure Reason	Device ID	Push Score	Suggested Push
UPI0206SF2_000	5/2/2026	Friday	15:01:49	15	PhonePe	CUST155944	25-34	Female	Karnataka	Mysuru	AcPiatt	Grocery	Online Shopping	Bank Account	ICICI Bank	152.17	0	0	Success	Android	61 No	0	0
UPI0206SF2_000	5/2/2026	Saturday	15:44:27	15	PhonePe	CUST103078	10-24	Female	Tamil Nadu	Madurai	AmazonPrime	Entertainment	Online Shopping	Bank Account	SBI	457.86	0	5.23	Success	Android	2 No	0	0
UPI0206SF2_000	5/10/2026	Thursday	13:09:13	13	PhonePe	CUST163783	10-24	Female	Rajasthan	Jodhpur	Digibasket	Grocery	P2M	Bank Account	Canara Bank	175.83	43.59	0	Success	Android	58 No	0	0
UPI0206SF2_000	5/8/2026	Thursday	23:21:05	21	PhonePe	CUST183070	35-44	Male	Telangana	Nizamabad	LocalRestaurant	Food&Dining	Bill Payment	Bank Account	Kotak Mahindra Bar	303.86	0	0	Success	iOS	32 No	0	0
UPI0206SF2_000	5/3/2026	Sunday	09:06:22	09	Paytm	CUST178072	25-34	Female	Delhi	Rohini	Zomato	Food&Dining	Bill Payment	UPI ID	Aus Bank	443.46	0	0	Success	Android	4 No	0	0
UPI0206SF2_000	5/6/2026	Friday	20:38:18	20	PhonePe	CUST418014	35-44	Male	MadhyaPradeh	Indore	Digibasket	Grocery	Bill Payment	Mobile Number	ICICI Bank	847.64	0	0	Success	Android	81 No	0	0
UPI0206SF2_000	5/2/2026	Thursday	19:46:23	19	Google Pay	CUST194307	25-34	Male	Delhi	Rohini	Loan EM	Financial Services	P2M	QR Scan	Kotak Mahindra Bar	823.72	0	0	Success	Android	68 No	0	0
UPI0206SF2_000	5/13/2026	Wednesday	16:53:24	16	Google Pay	CUST100489	35-44	Male	Maharashtra	Nashik	Swiggy	Food&Dining	P2M	QR Scan	Aus Bank	33.37	0	3.51	Success	iOS	96 No	0	0
UPI0206SF2_000	5/2/2026	Friday	09:18:53	09	PhonePe	CUST151546	45-54	Female	Gujarat	Rajkot	V	Recharge&Bills	Bill Payment	Mobile Number	ICICI Bank	421.61	0	0	Success	Android	71 No	0	0
UPI0206SF2_000	5/28/2026	Thursday	17:56:55	17	Google Pay	CUST1793878	35-44	Female	Gujarat	Surat	BroadbandBill	Utilities	P2M	UPI ID	SBI	380.97	0	0	Success	Android	55 No	0	0
UPI0206SF2_000	5/13/2026	Wednesday	13:12:45	13	Paytm	CUST103464	45-54	Male	Uttar Pradesh	Varanasi	Job	Recharge&Bills	P2P	UPI ID	HDFC Bank	153.65	57.01	0	Success	Android	8 No	0	0
UPI0206SF2_000	5/13/2026	Tuesday	09:42:14	09	PhonePe	CUST183873	35-44	Female	Tamil Nadu	Madurai	HP Field Pump	Fuel	Bill Payment	Mobile Number	SBI	704.4	0	0	Success	iOS	88 No	0	0
UPI0206SF2_000	5/13/2026	Tuesday	19:05:54	19	PhonePe	CUST103746	25-34	Male	Maharashtra	Pune	BroadbandBill	Utilities	Recharge	Bank Account	HDFC Bank	504.39	0	0	Success	Android	51 No	0	0
UPI0206SF2_000	5/18/2026	Tuesday	11:06:24	11	Paytm	CUST103072	35-44	Male	Delhi	Salari	Digibasket	Grocery	Recharge	Mobile Number	Kotak Mahindra Bar	105.55	0	5.55	Success	Android	76 No	0	0
UPI0206SF2_000	5/13/2026	Tuesday	12:40:53	12	Google Pay	CUST194056	10-24	Male	MadhyaPradeh	Jabalpur	Mutual Fund SIP	Financial Services	Bill Payment	UPI ID	Canara Bank	686.08	28.67	0	Success	Android	32 No	0	0
UPI0206SF2_000	5/3/2026	Saturday	12:23:22	12	Amazon Pay	CUST179396	35-44	Female	Telangana	Hyderabad	JobMart	Grocery	P2M	QR Scan	HDFC Bank	1423.41	4.58	0	Success	Android	100 No	0	0
UPI0206SF2_000	5/25/2026	Monday	11:17:44	11	PhonePe	CUST140440	25-34	Female	Gujarat	Rajkot	Myntra	Shopping	P2M	UPI ID	Punjab National Bar	378.85	0	0	Success	Android	36 No	0	0
UPI0206SF2_000	5/8/2026	Friday	21:41:04	21	PhonePe	CUST193664	35-44	Male	West Bengal	Howrah	Tata Play	Recharge&Bills	Bill Payment	UPI ID	SBI	419.25	72.41	0	Success	Android	80 No	0	0
UPI0206SF2_000	5/23/2026	Saturday	17:10:34	17	Google Pay	CUST130336	10-24	Female	Delhi	New Delhi	Insurance Premium	Financial Services	P2P	UPI ID	HDFC Bank	367.86	0	0	Success	Android	13 No	0	0
UPI0206SF2_000	5/9/2026	Saturday	19:10:28	19	Paytm	CUST1148791	25-34	Male	Uttar Pradesh	Lucknow	Gas Bill	Utilities	P2M	Mobile Number	Bank of Baroda	322.5	0	0	Success	Android	28 No	0	0
UPI0206SF2_000	5/16/2026	Friday	13:53:40	13	PhonePe	CUST141936	45-54	Male	Delhi	Rohini	BroadbandBill	Utilities	Online Shopping	Mobile Number	Aus Bank	637.36	0	0	Success	iOS	55 No	0	0
UPI0206SF2_000	5/18/2026	Sunday	18:14:43	18	PhonePe	CUST1447191	25-34	Female	Gujarat	Vadodra	Cla	Travel	P2M	UPI ID	Punjab National Bar	728.27	0	0	Success	Android	17 No	0	0
UPI0206SF2_000	5/16/2026	Friday	13:49:13	13	PhonePe	CUST103864	45-54	Male	Tamil Nadu	Chennai	Loan EM	Financial Services	Online Shopping	QR Scan	HDFC Bank	631.83	0	6.05	Success	Android	47 No	0	0
UPI0206SF2_000	5/28/2026	Thursday	08:52:02	08	Google Pay	CUST173789	25-34	Female	Karnataka	Mysuru	Bharat Petroleum	Fuel	P2P	QR Scan	Aus Bank	631.83	0	0	Success	Android	62 No	0	0
UPI0206SF2_000	5/4/2026	Monday	19:25:17	19	PhonePe	CUST1554301	25-34	Male	Rajasthan	Jodhpur	MalayTyp	Travel	Bill Payment	QR Scan	ICICI Bank	5535.53	0	0	Success	Android	32 No	0	0
UPI0206SF2_000	5/28/2026	Tuesday	08:58:36	08	Google Pay	CUST1323758	35-44	Female	Tamil Nadu	Salem	MSEB	Recharge&Bills	P2P	UPI ID	Canara Bank	126.66	0	0	Success	Android	66 No	0	0
UPI0206SF2_000	5/14/2026	Thursday	20:10:13	20	Amazon Pay	CUST194501	35-44	Female	Telangana	Hyderabad	Gas Bill	Utilities	P2P	Mobile Number	Punjab National Bar	336.18	0	0	Success	Android	37 No	0	0
UPI0206SF2_000	5/2/2026	Wednesday	18:35:36	18	Paytm	CUST183386	10-24	Female	Uttar Pradesh	Kanpur	Bharat Petroleum	Fuel	P2M	Mobile Number	Canara Bank	100.71	0	0	Success	Android	28 No	0	0
UPI0206SF2_000	5/6/2026	Friday	20:23:09	20	PhonePe	CUST1723777	35-44	Male	Gujarat	Surat	Reliance Digital	Shopping	P2M	UPI ID	Kotak Mahindra Bar	963.64	0	6.53	Success	Android	22 No	0	0
UPI0206SF2_000	5/6/2026	Friday	15:26:47	15	Google Pay	CUST1775545	55+	Female	Maharashtra	Nagpur	RCCTC	Travel	P2P	UPI ID	Punjab National Bar	321.26	70.87	0	Success	iOS	8 No	0	0
UPI0206SF2_000	5/23/2026	Friday	17:16:47	17	PhonePe	CUST106609	25-34	Male	Uttar Pradesh	Varanasi	Reliance Digital	Shopping	P2M	Mobile Number	Kotak Mahindra Bar	750.72	0	0	Success	Android	40 No	0	0
UPI0206SF2_000	5/1/2026	Thursday	17:57:37	17	CredPay	CUST132554	10-24	Male	Maharashtra	Nashik	Cla	Travel	Recharge	UPI ID	Punjab National Bar	533.15	65.01	0	Success	Android	27 No	0	0
UPI0206SF2_000	5/1/2026	Thursday	16:14:08	14	Google Pay	CUST1778274	35-44	Female	Delhi	Okhla	Cafe Coffee Day	Food&Dining	P2M	Bank Account	ICICI Bank	528.63	0	6.58	Success	Android	17 No	0	0
UPI0206SF2_000	5/1/2026	Wednesday	16:05:48	16	Paytm	CUST1302798	25-34	Male	Tamil Nadu	Madurai	BroadbandBill	Utilities	P2M	UPI ID	Canara Bank	435.8	0	0	Success	Android	42 No	0	0
UPI0206SF2_000	5/1/2026	Monday	09:07:09	09	PhonePe	CUST194954	45-54	Male	Karnataka	Hubbali	Swiggy	Food&Dining	P2M	UPI ID	Canara Bank	268.14	0	0	Success	Android	46 No	0	0
UPI0206SF2_000	5/1/2026	Monday	17:50:43	17	PhonePe	CUST161932	35-44	Female	Delhi	Okhla	LocalCafe	Healthcare	P2M	UPI ID	Aus Bank	45.26	0	0	Success	iOS	43 No	0	0
UPI0206SF2_000	5/4/2026	Monday	14:04:24	14	PhonePe	CUST170415	35-44	Male	West Bengal	Kolkata	Electronic Board	Recharge&Bills	Subscription	Mobile Number	HDFC Bank	138.6	0	2.58	Success	Android	3 No	0	0
UPI0206SF2_000	5/1/2026	Monday	13:24:13	13	PhonePe	CUST1807374	25-34	Female	Karnataka	Mangaluru	Tata Play	Recharge&Bills	P2M	QR Scan	HDFC Bank	246.13	0	0	Success	Android	10 No	0	0
UPI0206SF2_000	5/17/2026	Monday	12:35:55	12	Paytm	CUST1873091	10-24	Female	Karnataka	Mangaluru	BroadbandBill	Utilities	P2M	QR Scan	Aus Bank	628.33	0	2.1	Failed - Transaction Timeout	Android	26 No	0	0
UPI0206SF2_000	5/14/2026	Thursday	21:43:48	21	Amazon Pay	CUST192317	10-24	Male	Tamil Nadu	Coimbatore	Netflix	Entertainment	Bill Payment	QR Scan	HDFC Bank	172.82	0	0	Success	iOS	33 No	0	0
UPI0206SF2_000	5/16/2026	Friday	03:56:34	03	PhonePe	CUST193921	10-24	Female	MadhyaPradeh	Jabalpur	Domino's	Food&Dining	Value Transfer	Mobile Number	Aus Bank	1891.12	0	0	Success	Android	33 No	0	0
UPI0206SF2_000	5/17/2026	Sunday	19:49:41	19	PhonePe	CUST1245483	25-34	Female	Gujarat	Ahmedabad	Credit Card Bill	Financial Services	Bill Payment	Collect Request	ICICI Bank	107.66	0	5.01	Success	Android	62 No	0	0
UPI0206SF2_000	5/1/2026	Monday	13:11:27	13	Google Pay	CUST125401	10-24	Male	Uttar Pradesh	Vareasi	Uttar Pradesh	Education	P2M	QR Scan	Kotak Mahindra Bar	8662.19	0	0	Success	Android	54 No	0	0
UPI0206SF2_000	5/9/2026	Saturday	10:00:46	10	PhonePe	CUST186202	25-34	Female	Uttar Pradesh	Varanasi	Unacademy	Education	P2M	UPI ID	Bank of Baroda	575.36	42.09	3.58	Success	Android	85 No	0	0
UPI0206SF2_000	5/25/2026	Monday	20:16:32	20	PhonePe	CUST1473261	45-54	Female	Telangana	Hyderabad	Phacoo	Healthcare	Bill Payment	QR Scan	HDFC Bank	51.31	31.58	0	Success	Android	40 No	0	0
UPI0206SF2_000	5/2/2026	Tuesday	19:32:58	19	PhonePe	CUST1828103	25-34	Male	West Bengal	Kolkata	RCCTC	Travel	P2M	Mobile Number	SBI	84.25	0	0	Success	Android	45 No	0	0
UPI0206SF2_000	5/14/2026	Thursday	09:43:19	09	Paytm	CUST1347782	10-24	Male	Tamil Nadu	Coimbatore	AmazonPrime	Entertainment	P2M	QR Scan	SBI	998.3	0	2.58	Failed - Beneficiary Bank Off	iOS	97 No	0	0
UPI0206SF2_000	5/17/2026	Sunday	20:16:36	20	Google Pay	CUST184784	25-34	Female	Maharashtra	Mumbai	Zomato	Food&Dining	P2M	UPI ID	SBI	854.46	0	0	Success	Android	34 No	0	0
UPI0206SF2_000	5/2/2026	Wednesday	11:23:48	11	Google Pay	CUST102010	10-24	Male	Karnataka	Bengaluru	Job	Recharge&Bills	Value Transfer	QR Scan	Canara Bank	685.83	0	7.75	Success	Android	68 No	0	0
UPI0206SF2_000	5/23/2026	Saturday	19:45:59	19	PhonePe	CUST105816	25-34	Female	Rajasthan	Udaipur	LocalKtana	Grocery	P2M	Bank Account	HDFC Bank	311.52	0	4.84	Success	iOS	38 No	0	0
UPI0206SF2_000	5/1/2026	Sunday	16:00:39	16	Google Pay	CUST181039	45-54	Male	Rajasthan	Jodhpur	MSEB	Recharge&Bills	P2M	Mobile Number	Aus Bank	743.71	62.53	0	Success	Android	87 No	0	0
UPI0206SF2_000	5/16/2026	Friday	13:01:07	13	Google Pay	CUST1363930	35-44	Male	Delhi	Rohini	MedPlus	Healthcare	Bill Payment	UPI ID	Aus Bank	643.23	0	5.84	Success	Android	82 No	0	0
UPI0206SF2_000	5/2/2026	Wednesday	21:56:53	21	CredPay	CUST1607398	35-44	Male	Rajasthan	Kota	MedPlus	Healthcare	Bill Payment	Bank Account	Kotak Mahindra Bar	426.79	0	0	Success	Android	14 No	0	0
UPI0206SF2_000	5/1/2026	Monday	18:11:03	18	PhonePe	CUST1731953	45-54	Male	West Bengal	Kolkata	Society/Maintenance	Utilities	Bill Payment	UPI ID	Kotak Mahindra Bar	756.93	0	0	Success	Android	14 No	0	0
UPI0206SF2_000	5/16/2026	Wednesday	13:06:05	13	PhonePe	CUST1347127	25-34	Male	MadhyaPradeh	Bhopal	Indian Oil	Fuel	Online Shopping	UPI ID	Aus Bank	2532.54	0	0	Success	Android	39 No	0	0

2. Business Problem

A large transaction dataset contains valuable information, but analyzing individual transaction records manually makes it difficult to identify important patterns.

The key analytical challenge is to understand:

- Which states generate the highest transaction amounts?
- Which banks contribute the highest transaction value?
- Which transaction types are most significant?

- Which UPI applications account for the largest transaction value?
- Which age groups contribute most to transaction amounts?
- How do transaction amounts vary across days and hours?
- What proportion of transactions are successful, failed or pending?
- How many transactions are marked as suspected fraud?
- How can all these insights be presented through an interactive and easy-to-understand dashboard?

The project addresses these questions by converting raw transaction data into an interactive Excel dashboard.

3. Objectives

The main objectives of this project are:

1. Analyze 200,000 UPI transaction records from May 2026.
 2. Examine transaction amount patterns across different states and cities.
 3. Analyze transaction behavior across different age groups and genders.
 4. Compare transaction amounts across banks and transaction types.
 5. Analyze the contribution of different UPI applications.
 6. Evaluate successful, failed and pending transaction percentages.
 7. Analyze transaction amount trends by day and hour.
 8. Examine suspected fraudulent transactions using the existing Is_Suspected field.
 9. Build interactive dashboards using Excel PivotTables and slicers.
 10. Convert the analysis into meaningful business insights and recommendations.
-

4. Dataset Description

The project uses a sample UPI transaction dataset containing **200,000 transaction records** covering **May 1, 2026 to May 31, 2026**.

some rows were removed during the preparation process for smoothly run excel.

The dataset contains transaction, customer, geographical, merchant, payment, banking, financial and risk-related information.

4.1 Dataset Characteristics

Attribute	Description
Dataset type	UPI transaction data
Time period	May 1–31, 2026
Number of transactions	200,000
Data preparation	Day and Hour fields created

Attribute	Description
Source data cleaning	Dataset was already cleaned
Analysis tool	Microsoft Excel
Dashboard pages	2
Main analysis metric	Transaction Amount

5. Features and Terminologies

5.1 Main Features

Feature	Description
Transaction_ID	Unique identifier of a transaction
Transaction_Date	Date on which the transaction occurred
Day	Day of the week derived from the transaction date
Transaction_Time	Time at which the transaction occurred
Hour	Hour extracted from transaction time
UPI_App	UPI application used for the transaction
Customer_ID	Identifier of the customer
Age_Group	Customer age segment
Gender	Customer gender
State	State associated with the transaction
City	City associated with the transaction
Merchant_Name	Merchant associated with the transaction
Merchant_Category	Category of the merchant
Transaction_Type	Type of transaction
Payment_Method	Method used to make the payment
Bank_Name	Bank associated with the transaction
Amount	Transaction value in INR
Cashback	Cashback associated with the transaction
Transaction_Fee	Fee associated with the transaction
Transaction_Status	Success, Failed or Pending status
Failure_Reason	Reason associated with failed transactions
Device	Device type used
Risk_Score	Risk score associated with the transaction
Is_Suspected	Indicates whether a transaction is suspected

5.2 Important Terminologies

Transaction Amount:

The monetary value of a UPI transaction.

Success Rate:

The percentage of transactions that were successfully completed.

Cashback:

The cashback amount associated with transactions.

Transaction Type:

The category of transaction, such as P2P, P2M, Recharge, Bill Payment, Subscription, Online Shopping and Wallet Transfer.

UPI App:

The application through which the UPI transaction was performed.

Risk Score:

A score associated with the potential risk of a transaction.

Suspected Fraud:

Transactions marked as suspected using the existing Is_Suspected field.

6. Tools and Technologies

The project was developed using:

Microsoft Excel

Excel was used for:

- Data preparation
- Creating Day and Hour fields
- Calculations
- PivotTables
- Slicers
- KPI analysis
- Data visualization
- Dashboard development

The project uses Excel rather than a programming-based analytics stack because the main objective was to demonstrate spreadsheet-based data analysis and dashboard development.

7. Data Preparation

The source dataset was already cleaned, so no major cleaning operation or row removal was performed.

The main preparation activities included:

7.1 Creating Day

A Day field was created from the transaction date to enable weekday-level analysis.

The resulting categories include:

- Sunday
- Monday
- Tuesday
- Wednesday
- Thursday
- Friday
- Saturday

7.2 Creating Hour

The transaction time was used to create an Hour field ranging from:

00 to 23

This enabled transaction amount analysis across different hours of the day.

7.3 Existing Age Group

The Age_Group field was already available in the source dataset. It was used directly for demographic analysis.

The dashboard contains:

- 18–24
- 25–34
- 35–44
- 45–54
- 55+

7.4 Financial Presentation

Transaction amounts were presented in **Crore (Cr)** and **Lakh (L)** where appropriate to make large values easier to interpret.

The conversions used were:

- ₹1 Crore = ₹10,000,000
- ₹1 Lakh = ₹100,000

For example:

₹17.12 Crore represents approximately ₹17.12 crore in total transaction value.

8. Analytical Methodology

The analysis followed this workflow:

Raw Data → Prepared Data → Calculations → PivotTables → Slicers → Dashboard 1 & Dashboard 2 → Insights

8.1 PivotTable Analysis

PivotTables were used to summarize transaction data across different dimensions, including:

- State
- Bank
- Transaction Type
- UPI App
- Age Group
- Transaction Status
- Day
- Hour
- Date

The primary financial measure used for amount-based analysis was **Sum of Amount**.

8.2 Interactive Slicers

Slicers were added to allow users to dynamically filter the dashboard.

Dashboard 1 filters

- City
- Merchant Name

Dashboard 2 filters

- Merchant Category
- Gender

The slicers allow users to investigate specific segments without manually filtering the underlying data.

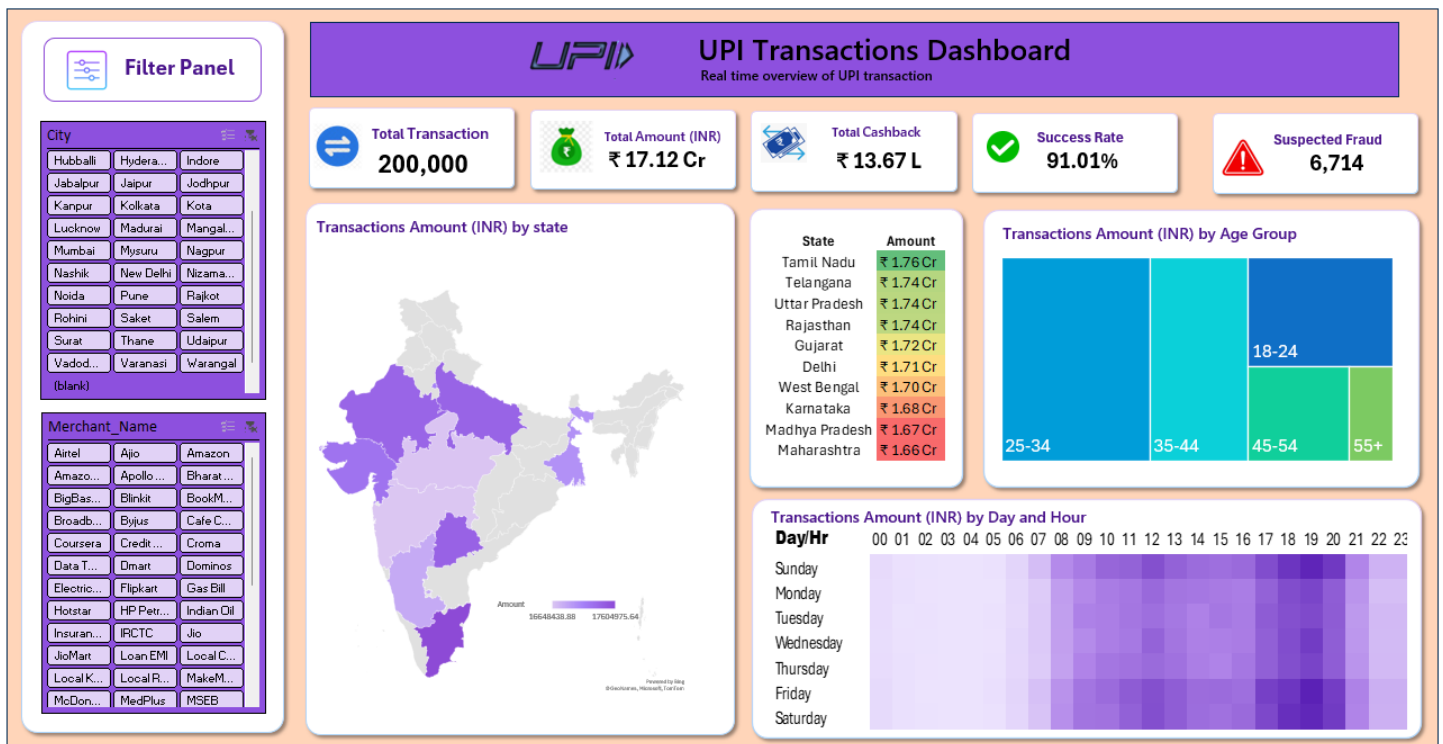
9. Dashboard Development

Two dashboard pages were developed to present the analysis from different perspectives.

10. Dashboard 1 — Geographic, Demographic & Time Analysis

Dashboard 1 focuses on geographical, demographic and time-based transaction analysis.

Figure 2: Dashboard 1 — Geographic, Demographic & Time Analysis



10.1 KPI Cards

The dashboard displays five major KPIs:

KPI	Result
Total Transactions	200,000
Total Amount	₹17.12 Cr
Total Cashback	₹13.67 L
Success Rate	91.01%
Suspected Fraud	6,714

Count of Transaction	Sum of Amount	Sum of Cashback_INR
200000	171242924.6	1367449.01
200,000	171,242,925	1,367,449
	₹ 17.12 Cr	₹ 13.67 L

These KPIs provide a quick overview of the dataset.

10.2 Transaction Amount by State

A map visual was used to display the geographical distribution of transaction amounts across Indian states.

The state ranking shown in the dashboard includes:

State	Transaction Amount
Tamil Nadu	₹1.76 Cr
Telangana	₹1.74 Cr
Uttar Pradesh	₹1.74 Cr
Rajasthan	₹1.74 Cr

State	Transaction Amount
Gujarat	₹1.72 Cr
Delhi	₹1.71 Cr
West Bengal	₹1.70 Cr
Karnataka	₹1.68 Cr
Madhya Pradesh	₹1.67 Cr
Maharashtra	₹1.66 Cr

State	Amount
Tamil Nadu	₹1.76 Cr
Telangana	₹1.74 Cr
Uttar Pradesh	₹1.74 Cr
Rajasthan	₹1.74 Cr
Gujarat	₹1.72 Cr
Delhi	₹1.71 Cr
West Bengal	₹1.70 Cr
Karnataka	₹1.68 Cr
Madhya Pradesh	₹1.67 Cr
Maharashtra	₹1.66 Cr

Tamil Nadu records the highest transaction amount among the states displayed in the dashboard.

10.3 Transaction Amount by Age Group

A treemap was used to compare transaction amounts across age groups.

The groups analyzed are:

- 18–24
- 25–34
- 35–44
- 45–54
- 55+

Age	amount
18-24	₹3.38 Cr
25-34	₹6.49 Cr
35-44	₹4.31 Cr
45-54	₹2.06 Cr
55+	₹.88 Cr

The visualization indicates that the **25–34 age group represents the largest segment by transaction amount** among the displayed age groups.

10.4 Transaction Amount by Day and Hour

A heatmap was used to analyze transaction amounts across:

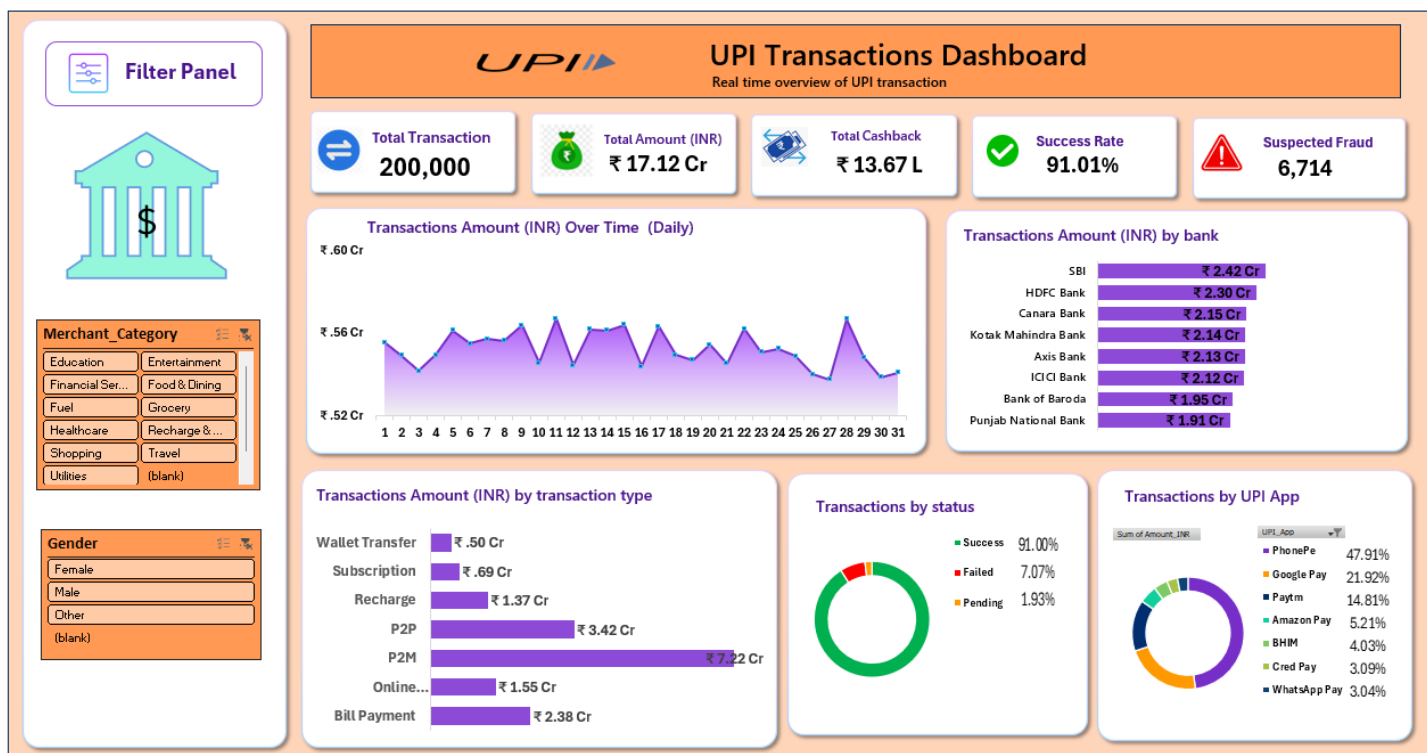
- Seven days of the week
- 24 hours of the day

This visualization makes it possible to identify periods with relatively higher or lower transaction activity and supports time-based operational analysis.

11. Dashboard 2 — Banking, Transaction & Payment Analysis

Dashboard 2 focuses on banking, transaction type, payment application and transaction-status analysis.

Figure 3: Dashboard 2 — Banking, Transaction & Payment Analysis



11.1 Daily Transaction Amount

A line chart displays transaction amount across the 31 days of May 2026.

The metric used is:

Sum of Amount per Day

This provides a view of how transaction value changes throughout the month.

11.2 Transaction Amount by Bank

The dashboard compares transaction amounts across major banks.

Bank	Transaction Amount
SBI	₹2.42 Cr
HDFC Bank	₹2.30 Cr
Canara Bank	₹2.15 Cr
Kotak Mahindra Bank	₹2.14 Cr
Axis Bank	₹2.13 Cr
ICICI Bank	₹2.12 Cr
Bank of Baroda	₹1.95 Cr
Punjab National Bank	₹1.91 Cr

SBI has the highest transaction amount among the banks displayed.

Punjab Nati	₹ 1.91 Cr
Bank of Bar	₹ 1.95 Cr
ICICI Bank	₹ 2.12 Cr
Axis Bank	₹ 2.13 Cr
Kotak Mahir	₹ 2.14 Cr
Canara Ban	₹ 2.15 Cr
HDFC Bank	₹ 2.30 Cr
SBI	₹ 2.42 Cr

11.3 Transaction Amount by Transaction Type

The dashboard shows:

Transaction Type	Amount
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Wallet Transfer	₹0.50 Cr
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Subscription	₹0.69 Cr
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Recharge	₹1.37 Cr
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P2P	₹3.42 Cr
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P2M	₹7.22 Cr
-----	----------

Online Shopping	₹1.55 Cr
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Bill Payment	₹2.38 Cr
--------------	----------

Bill Payment	₹ 2.38 Cr
Online Shopping	₹ 1.55 Cr
P2M	₹ 7.22 Cr
P2P	₹ 3.42 Cr
Recharge	₹ 1.37 Cr
Subscription	₹ .69 Cr
Wallet Transfer	₹ .50 Cr

P2M represents the largest transaction amount among the displayed transaction types.

11.4 Transaction Status

Transaction status is analyzed using transaction count.

Status	Percentage
--------	------------

Success	91.00%
---------	--------

Failed	7.07%
--------	-------

Pending	1.93%
---------	-------

Success	91.00%
Failed	7.07%
Pending	1.93%

The small difference between the 91.00% shown in the status chart and the 91.01% KPI is attributable to rounding/calculation presentation.

11.5 UPI Application Analysis

The UPI application analysis is based on **transaction amount**, rather than transaction count.

UPI Application	Share of Amount
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PhonePe	47.91%
---------	--------

Google Pay	21.92%
------------	--------

Paytm	14.81%
-------	--------

Amazon Pay	5.21%
------------	-------

BHIM	4.03%
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Cred Pay	3.09%
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WhatsApp Pay	3.04%
--------------	-------

PhonePe	47.91%
Google Pay	21.92%
Paytm	14.81%
Amazon Pay	5.21%
BHIM	4.03%
Cred Pay	3.09%
WhatsApp Pay	3.04%

PhonePe has the largest share of transaction amount among the applications shown.

12. Key KPI Results

The overall dashboard provides the following results:

Total Transactions — 200,000

The dataset contains 200,000 transaction records for May 2026.

Total Transaction Amount — ₹17.12 Cr

The total transaction value represented in the dataset is ₹17.12 crore.

Total Cashback — ₹13.67 L

The transactions collectively account for ₹13.67 lakh in cashback.

Success Rate — 91.01%

Approximately 91% of transactions are successful, indicating that successful transactions make up the dominant share of transaction records.

Suspected Fraud — 6,714

A total of 6,714 transactions are marked as suspected using the existing Is_Suspected field.

13. Key Insights

13.1 Geographic Insight

Tamil Nadu has the highest transaction amount among the states shown, at approximately **₹1.76 crore**.

This indicates a relatively high transaction value contribution from Tamil Nadu within the analyzed sample.

13.2 Demographic Insight

The **25–34 age group** represents the largest segment in the age-group transaction amount analysis.

This suggests that younger working-age customers represent a major portion of the transaction value in the dataset.

13.3 Banking Insight

SBI records the highest transaction amount among the banks displayed, with approximately **₹2.42 crore**.

HDFC Bank follows with approximately ₹2.30 crore.

13.4 Transaction Type Insight

P2M is the largest transaction category by amount, with approximately **₹7.22 crore**.

This is substantially higher than the other transaction categories displayed and indicates the importance of merchant-related payments within the dataset.

13.5 UPI Application Insight

PhonePe has the largest share of transaction amount at **47.91%**, followed by Google Pay at **21.92%** and Paytm at **14.81%**.

This indicates a strong concentration of transaction value among the leading UPI applications.

13.6 Transaction Status Insight

Successful transactions account for approximately **91%** of transaction records, while failed transactions represent **7.07%** and pending transactions represent **1.93%**.

Although successful transactions dominate, the failed transaction percentage is still significant enough to justify monitoring transaction failures and their underlying reasons.

13.7 Time-Based Insight

The Day/Hour heatmap provides a time-based view of transaction value across the month.

This type of analysis can help identify periods of relatively high transaction activity and potentially support operational planning.

14. Business Recommendations

Based on the analysis, the following recommendations can be considered:

14.1 Monitor High-Value Transaction Channels

Banks and payment providers can monitor high-volume banking channels to ensure reliable transaction processing during periods of high activity.

14.2 Improve Failed Transactions

With failed transactions accounting for approximately 7.07% of transaction records, organizations can analyze failure reasons and identify opportunities to improve transaction reliability.

14.3 Monitor Fraud Indicators

The 6,714 suspected transactions should be investigated using risk scores, transaction characteristics, device information, location and other available attributes.

The Is_Suspected field should be treated as an indicator rather than proof of actual fraud.

14.4 Optimize UPI Application Experience

The strong transaction-value contribution from PhonePe, Google Pay and Paytm suggests that payment providers and merchants should prioritize a smooth experience across the most widely used applications.

14.5 Use Time-Based Monitoring

Day and hour analysis can help organizations identify high-activity periods and plan infrastructure capacity and monitoring resources accordingly.

14.6 Segment Customer Analysis

The age-group analysis can be combined with gender, location and merchant category to develop more detailed customer behavior segments.

15. Limitations

This project has several limitations that should be acknowledged:

1. The analysis is based on a sample dataset rather than live UPI transaction data.
 2. The dataset was obtained from a publicly available YouTube tutorial/sample.
 3. The analysis covers May 2026 only.
 4. Suspected fraud is based on the existing `Is_Suspected` field and does not represent independently verified fraud.
 5. The dashboard was developed in Microsoft Excel and is not connected to a live transaction system.
 6. The analysis identifies patterns and relationships but does not establish causal relationships.
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16. Conclusion

The UPI Transaction Analytics Dashboard transforms 200,000 transaction records from May 2026 into an interactive analytical solution using Microsoft Excel.

The project demonstrates the use of data preparation, calculated analysis, PivotTables, slicers and dashboard visualizations to analyze transaction behavior from multiple perspectives.

The analysis identifies important patterns across states, age groups, banks, transaction types, UPI applications, transaction statuses and time periods. Key findings include a total transaction value of ₹17.12 crore, a 91.01% overall success rate, 6,714 suspected transactions, a high contribution from P2M transactions, and a strong transaction-value share for PhonePe.

Overall, the project demonstrates how Excel can be used to convert a large transaction dataset into an interactive dashboard that supports exploratory analysis and business-oriented decision-making.

17. Future Scope

The project can be extended in several ways:

- Connect the dashboard to regularly updated transaction data.

- Build automated data-refresh pipelines.
 - Develop machine-learning-based fraud detection.
 - Predict future transaction amounts using time-series forecasting.
 - Perform customer segmentation using clustering.
 - Analyze transaction failure reasons in greater detail.
 - Develop advanced geographic analysis.
 - Integrate the dashboard with Power BI for scalable reporting.
 - Add automated alerts for high-risk or unusual transactions.
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18. Project Workflow Summary

The overall workflow can be summarized as:

Source Dataset ↓

Raw Data ↓

Prepared Data ↓

Create Day & Hour ↓

Excel Calculations ↓

PivotTables ↓

Slicers ↓

Dashboard 1 ↓

Dashboard 2 ↓

KPI Analysis ↓

Business Insights & Recommendations

19. Final Project Statement

This project demonstrates practical data analytics skills using Microsoft Excel, including data preparation, calculated analysis, PivotTables, slicers, KPI development and interactive dashboard design. It focuses on converting transaction-level data into business-oriented insights rather than simply presenting raw numbers.
